

Seeing the Signal

Image-Based Deep Learning for Limit Order Book Price Prediction

Candidates:     

What's a limit order book?

Think of it as a **live scoreboard** of who wants to buy and sell a stock, at what price, and how much.

- Asks (sellers) in red, going up
- Bids (buyers) in green, going down
- 10 levels each side = 40 features per snapshot
- $\text{Mid-price} = (\text{best ask} + \text{best bid}) / 2$

Ask 10 \$20.69 45

Ask 5 \$20.64 90

Ask 1 \$20.60 150

--- mid \$20.55 ---

Bid 1 \$20.50 200

Bid 5 \$20.46 110

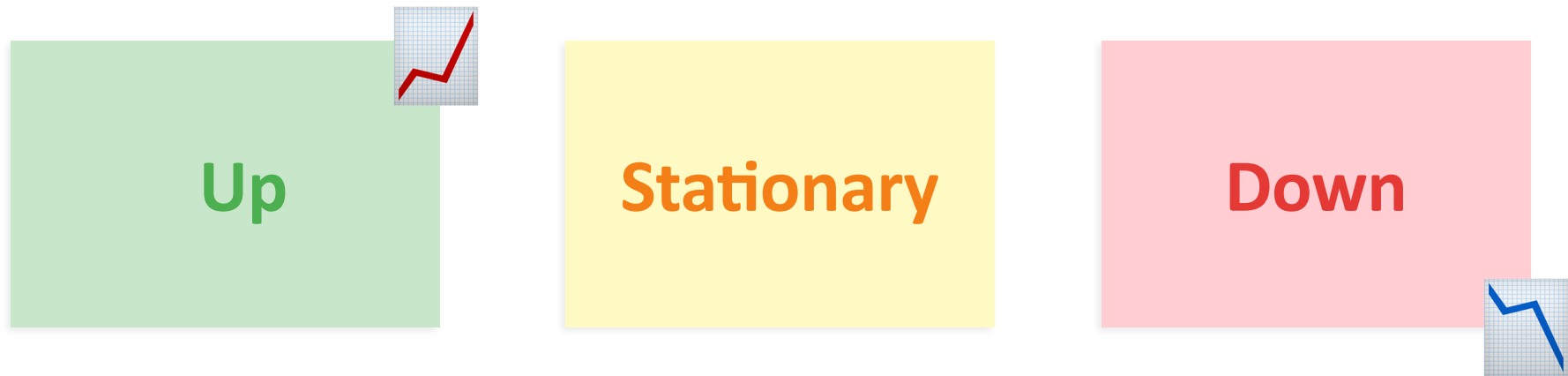
Bid 10 \$20.41 80

*^ this updates thousands
of times per second*

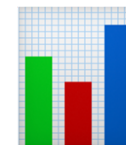
Our aim



Given the last 100 snapshots, predict what happens to the mid-price:



Class balance: 34.4% up | 31.7% stationary | 33.9% down



The dataset: FI-2010

The standard benchmark everyone compares against (Ntakaris et al., 2018)

- 5 Finnish stocks on Nasdaq Nordic, 10 consecutive trading days (2010)
- Roughly 4.5 million limit order events, downsampled to every 10 events
- Setup 2: train on days 1-7, test on days 8-10 (following Zhang et al. (2019))
- Decimal precision normalization
- Labels: smoothed future mid-price vs. current, thresholded into 3 classes
- At k=50, training set is approximately balanced: 34.4% Up, 31.7% Stationary, 33.9% Down



Known limitation: FI-2010 contains only 10 days of relatively illiquid stocks. Models tuned on it tend to lose performance when transferred to more liquid markets like NASDAQ (Prata et al., 2023).

Previous work



Classical ML

SVM and multilinear discriminant methods using handcrafted features achieved F1 scores of 35-48% at k=10 on FI-2010 (Zhang et al., 2019)

Neural networks

DeepLOB (Zhang et al., 2019) is the current best at 80.35% weighted F1 (k=50). It uses custom (1x2) filters engineered to pair prices with volumes, plus an Inception module and LSTM. All prior neural models treat the LOB as a flat numerical grid.

Image-based

Nino et al. (2018) encoded LOB data as images and achieved 74% directional accuracy, but used a basic CNN and did not benchmark on FI-2010. Backhouse et al. (2025) used LOB images for generation, not prediction.

The gap: image-based LOB representations have never been benchmarked against established models on FI-2010.

Why images?



It already looks like one

The LOB has price levels along one axis and time along another, with volume as intensity. This is a heatmap by nature, and CNNs are designed for exactly this kind of spatially structured input.



Patterns are local

Predictive signals like bid-ask imbalance and volume spikes occur across nearby price levels and adjacent timesteps. A CNN's local receptive field is well matched to detect these.



Translation invariance

A pattern appearing at $t=10$ or $t=80$ carries the same signal. CNNs reuse the same learned filters across the image via weight sharing, so they handle this naturally.

LOB-to-Image Pipeline



1

Sliding Window

Extract 100 consecutive timesteps from the raw LOB features.

Shape: (100, 40)

>

2

Reshape

Rearrange into 4 channels of 10 levels x 100 timesteps.

Stored as (10, 100, 4), then permuted to (4, 10, 100) for the CNN.

>

3

Channel Split

Ch 0: Bid volume

Ch 1: Bid price (centred)

Ch 2: Ask volume (negated)

Ch 3: Ask price (centred)

Ask volumes negated so bid/ask differ in sign.

>

4

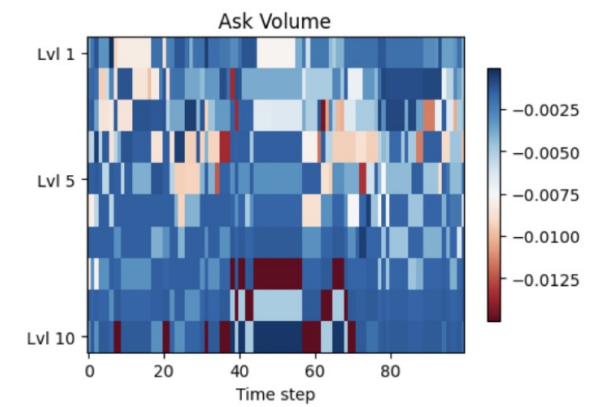
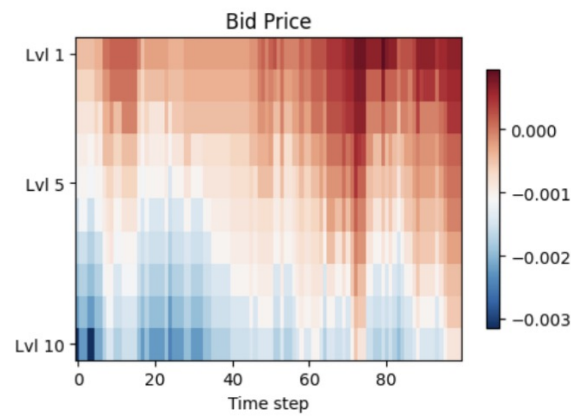
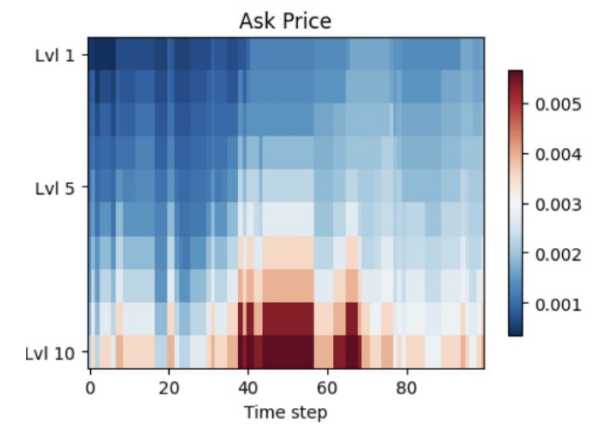
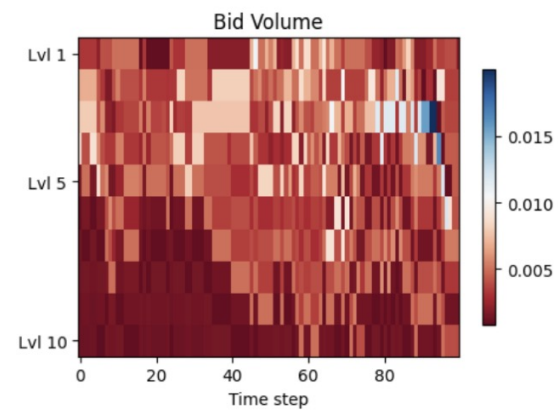
Centring

Both price channels are centred on the window's opening mid-price:

How to read the image:

- Height axis: 10 price levels per channel (level 1 = best bid/ask, level 10 = deepest)
- Width axis: 100 consecutive timestamps (left = oldest, right = most recent)
- Volumes: Positive for bids, negated for asks. Prices: deviations from opening mid-price.

The images





What we built

Each model answers a different question:

LDA / LinearSVC

Can a linear classifier on raw features do anything at all?

CNN-MLP

Does the image alone carry signal? (no temporal model)

CNN-LSTM

Does modelling temporal dependencies help?

DeepLOB

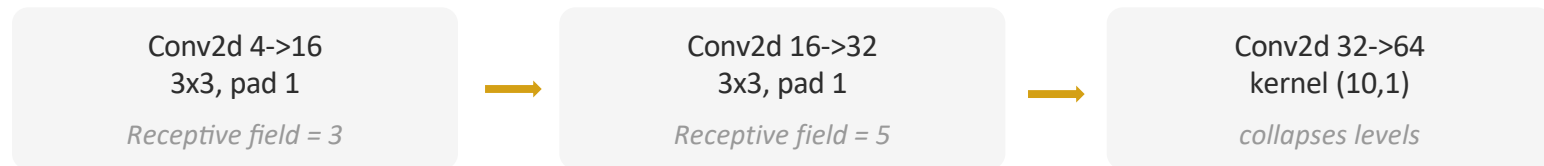
Can our painting match a purpose-built architecture?

CNN-MLP: ~845K params | CNN-LSTM: ~57K params | DeepLOB: ~60K params (reported)

Model Architectures

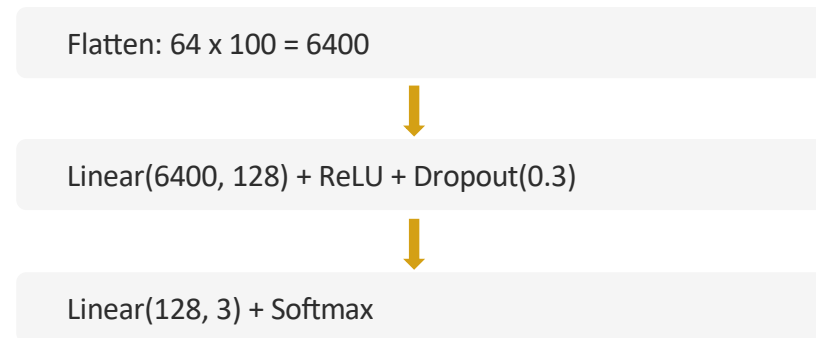


Shared CNN encoder



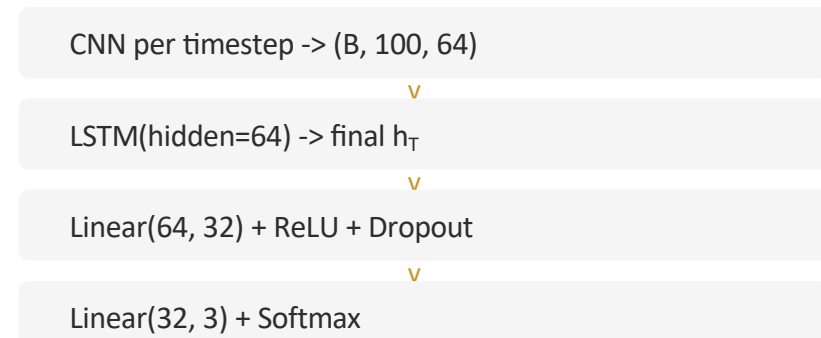
No pooling. Stride 1 throughout. Top levels carry more signal, so we preserve full spatial resolution.

CNN-MLP



The CNN sees the whole 100-step window as a single image.

CNN-LSTM



LSTM processes the 100-step sequence and uses gating for long-range dependencies.



Hyperparameter tuning & training

Two-stage grid search, both stages using AdamW with cross-entropy loss:

- Stage 1: learning rate {5e-4, 1e-3, 2e-3} x batch size {32, 64, 128} = 9 trials per model
- Stage 2: fix Stage 1 winner, search dropout {0.0, 0.1, 0.2, 0.3, 0.5} x weight decay {0, 1e-5, 1e-4, 5e-4, 1e-3} = 25 trials per model
- Ranked by best validation weighted F1, with validation loss as tiebreaker

Selected configurations for final training:

	CNN-MLP	CNN-LSTM
Learning rate	0.001	0.002
Batch size	32	64
Dropout	0.3	0.0
Weight decay	0	0.0001
Max epochs	30	30
Early stopping	Patience 7	Patience 7

All runs: seed 42 | single NVIDIA T4 GPU | AdamW optimiser | softmax cross-entropy loss

Results

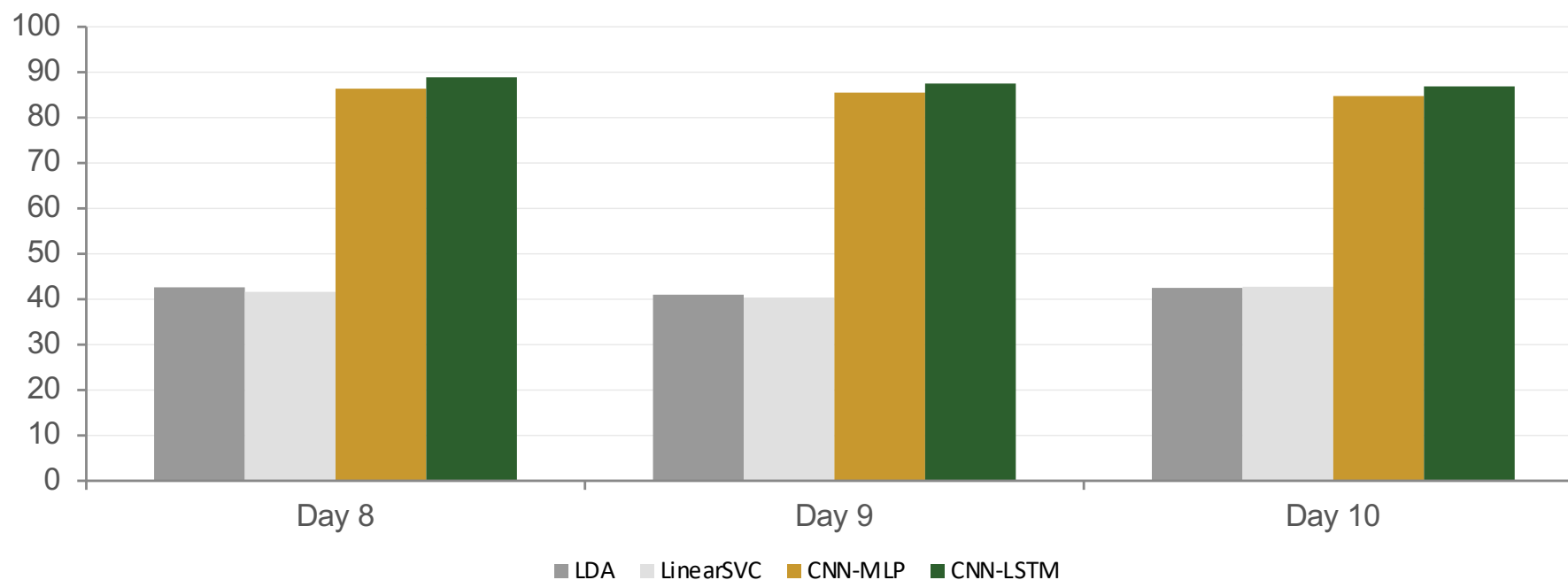


Model	Accuracy	Precision	Recall	F1 (wtd)
LDA	41.63	44.95	41.63	42.25
LinearSVC	41.11	44.80	41.11	41.70
DeepLOB	80.51	80.38	80.51	80.35
CNN-MLP	85.75	85.79	85.75	85.70
CNN-LSTM	88.02	88.36	88.02	87.93

CNN-MLP beats DeepLOB by +5.35 F1 points with no temporal model. CNN-LSTM adds a further +2.23 F1.

- Most of the gain comes from the image encoding itself. The LSTM adds a smaller temporal boost on top.
- Non-neural baselines (~42 F1) confirm the signal is in cross-level, cross-time non-linear patterns.

Per-day consistency



Both image-based models stay consistent across unseen test days. Within-model variation never exceeds 3 F1 points, suggesting the learned representations generalise across days rather than overfitting to day-specific patterns.

Conclusion



By encoding LOB snapshots as four-channel images, standard CNNs outperform DeepLOB on FI-2010 without any specialised architecture. The predictive structure of the order book is largely spatial.

87.93%

CNN-LSTM F1

+7.58 vs DeepLOB

85.70%

CNN-MLP F1

no temporal model

80.35%

DeepLOB F1

purpose-built



Limitations

- FI-2010 is only 10 days of relatively illiquid Nordic stocks from 2010. Models tuned on it lose substantial performance on more recent, liquid markets (Prata et al., 2023).
- We evaluated at a single prediction horizon (k=50). Testing across horizons would reveal whether the gain over DeepLOB is stable or specific to k=50.
- Results are from single training runs per configuration. We cannot quantify variance across seeds.
- The reported gains are evidence on this specific benchmark, not a general claim.



Future work

- Vision transformers (ViT-LOB, Liu et al., 2024)
- Dual attention over features + time (TLOB, Berti & Kasneci, 2025): replaces the CNN-LSTM stack with a single model that learns spatial and temporal patterns jointly.
- More recent and more liquid order book data (e.g. NASDAQ) to test whether the image representation advantage transfers.
- Alternative image encodings and fusion models combining image and numerical input.

Bibliography

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